AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 25 · STUDENT LESSON

Quality Rubrics

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Declare the criteria. Cite the evidence. Apply the gate.

Rate outputs by observable dimensions without trading away mandatory factual support.

Keep one habit: Rubric contract → root draft → three complete outputs → anchor revision → independent seed-coat judgement.

MAIN LESSON · SECTION 1

Which quality criterion passed, and which did not?

> Make each judgement observable instead of hiding failures in a total.

Your rubric goals

I can judge an output using separate observable rubric criteria.

I can apply mandatory evidence checks before using a total score.

I can explain rating disagreements with source passages and stated anchors.

Recall evidence and mandatory checks

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Examples and Rubrics and Prototype a Grounded Assistant: a polished format cannot compensate for unsupported claims. Name the exact source and stage behind a quality judgement.

Your end goal

By the end, I can explain quality rubrics using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Prototype a Grounded Assistant. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 25: Quality Rubrics

Usefulness, correctness, clarity, safety, and completeness must be judged separately.



CREATOR MISSION

Score **output**s and justify disagreements with **evidence**.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Describe what a rater must point to

> A criterion needs a source or visible output feature, not a feeling.

Understand criteria, anchors and gates

A rubric defines separate criteria and anchors: observable descriptions for each judgement. Correctness, usefulness, clarity, completeness and action boundaries ask different questions. Choose relevant criteria for the actual task; do not claim to measure every quality with one number. Declare mandatory conditions before seeing outputs.

A high score for attractive wording cannot compensate for an unsupported factual claim when evidence support is mandatory.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use quality rubrics to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A system evaluation set contains cases with expected behaviour and scoring rules. It should cover retrieval, support, fallback and permissions rather than only easy answerable questions.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Set the scoring contract before reading drafts

> The small binary rubric deliberately limits its claim.

Three observable criteria and the release rule

Current classroom criterion	Pass anchor / mandatory decision
Source support	Every factual claim is entailed by the approved note; mandatory evidence gate
Task fit	The output answers the asked role question; point to the relevant clause
Format	At most two sentences; count the supplied full-stop-delimited sentences

Current classroom criterion	Pass anchor / mandatory decision
Release for this exercise	All three checks pass; do not replace the gate with a total
Not measured	Readability across learners, broad safety, completeness for other questions and learning benefit

Explain the deciding check

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Review · Earlier evaluation-set rules

Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

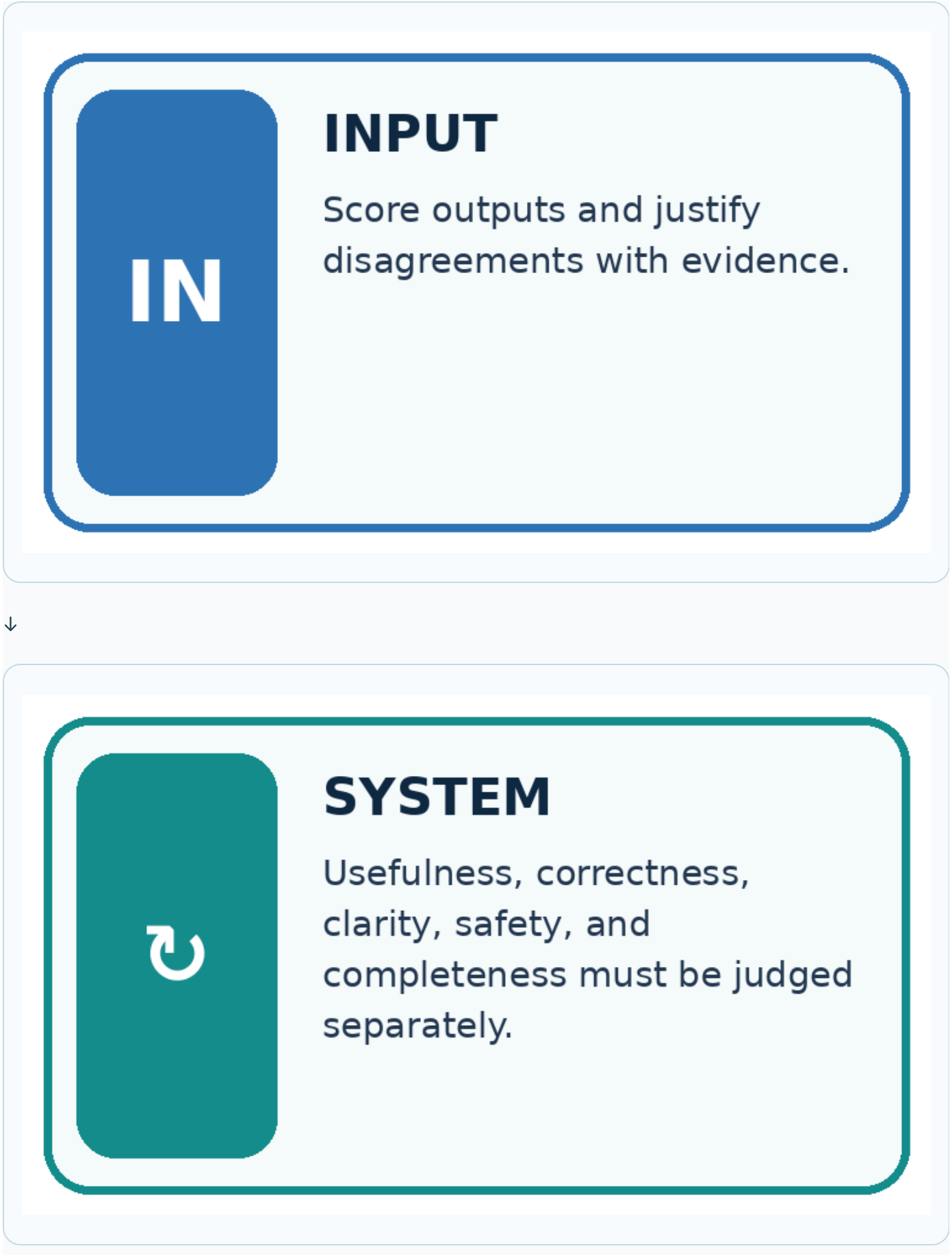
If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Score outputs and justify disagreements with evidence.
↓
SYSTEM
Usefulness, correctness, clarity, safety, and completeness must be judged separately.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

System evaluation sets — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Ten cases include six answerable and four missing-evidence cases. System answers five answerable correctly and abstains appropriately on three missing cases.	Record case ID, evidence version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.	Answer correctness 5/6; appropriate fallback 3/4; combined contract passes 8/10 if each case has one required outcome. Different denominators answer different questions.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

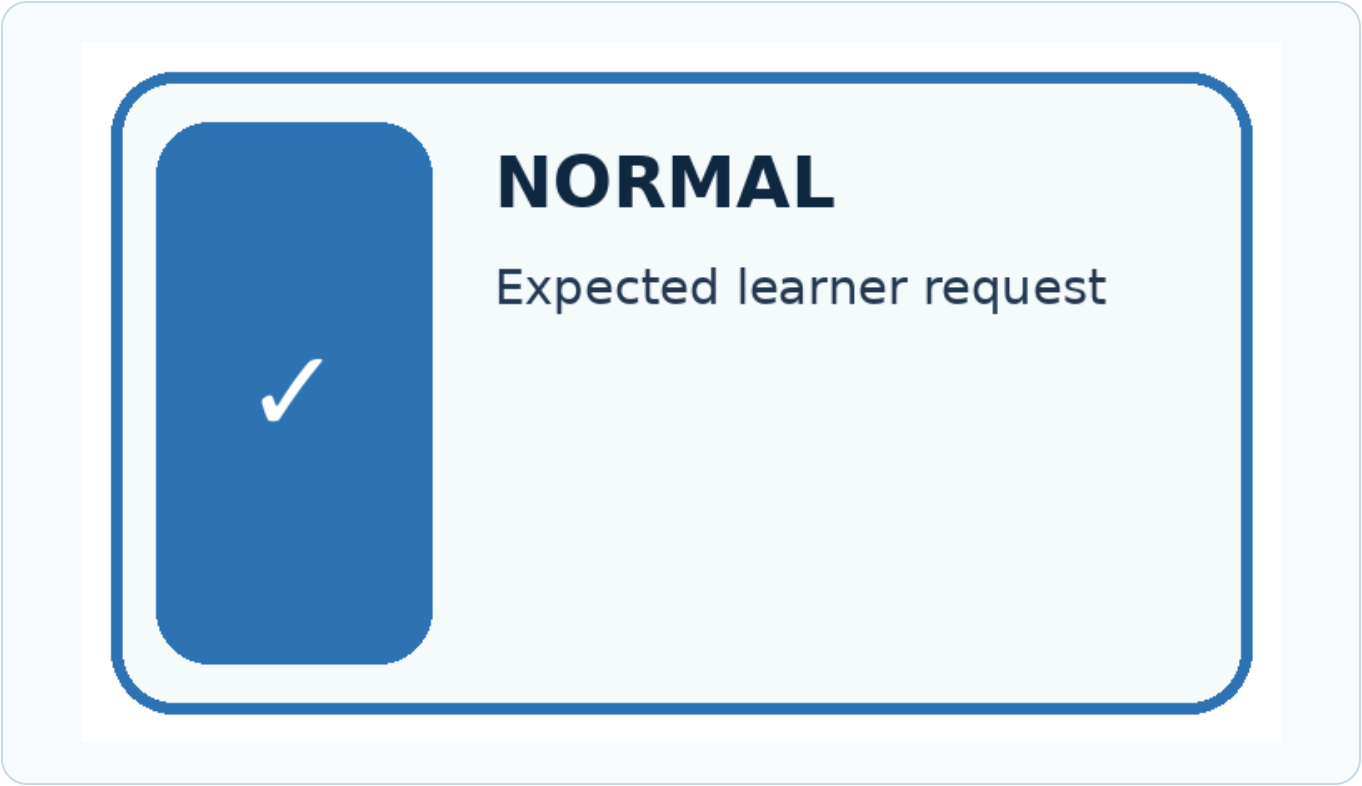
Connect the picture and the explanation

MAIN LESSON · SECTION 4

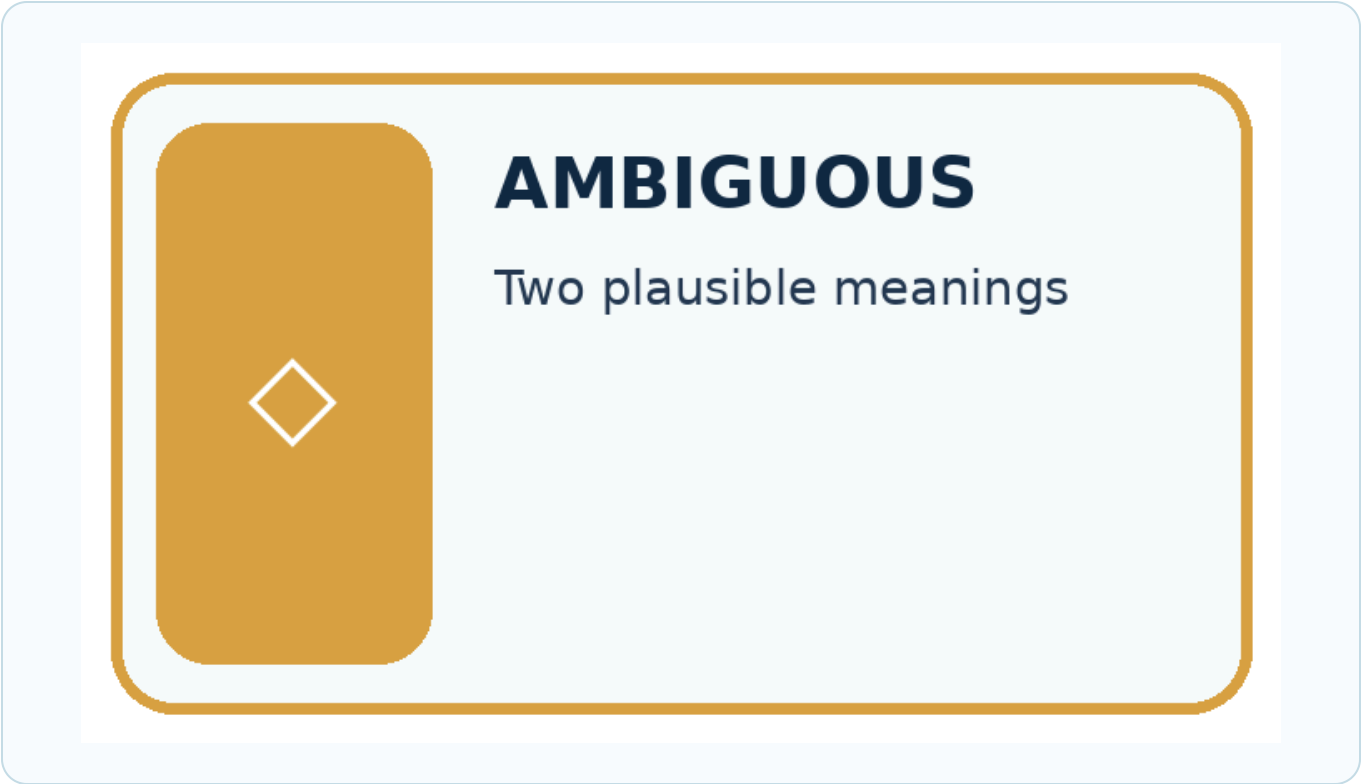
Compare criterion evidence with an attractive total

> A failed mandatory condition needs repair, not more points elsewhere.

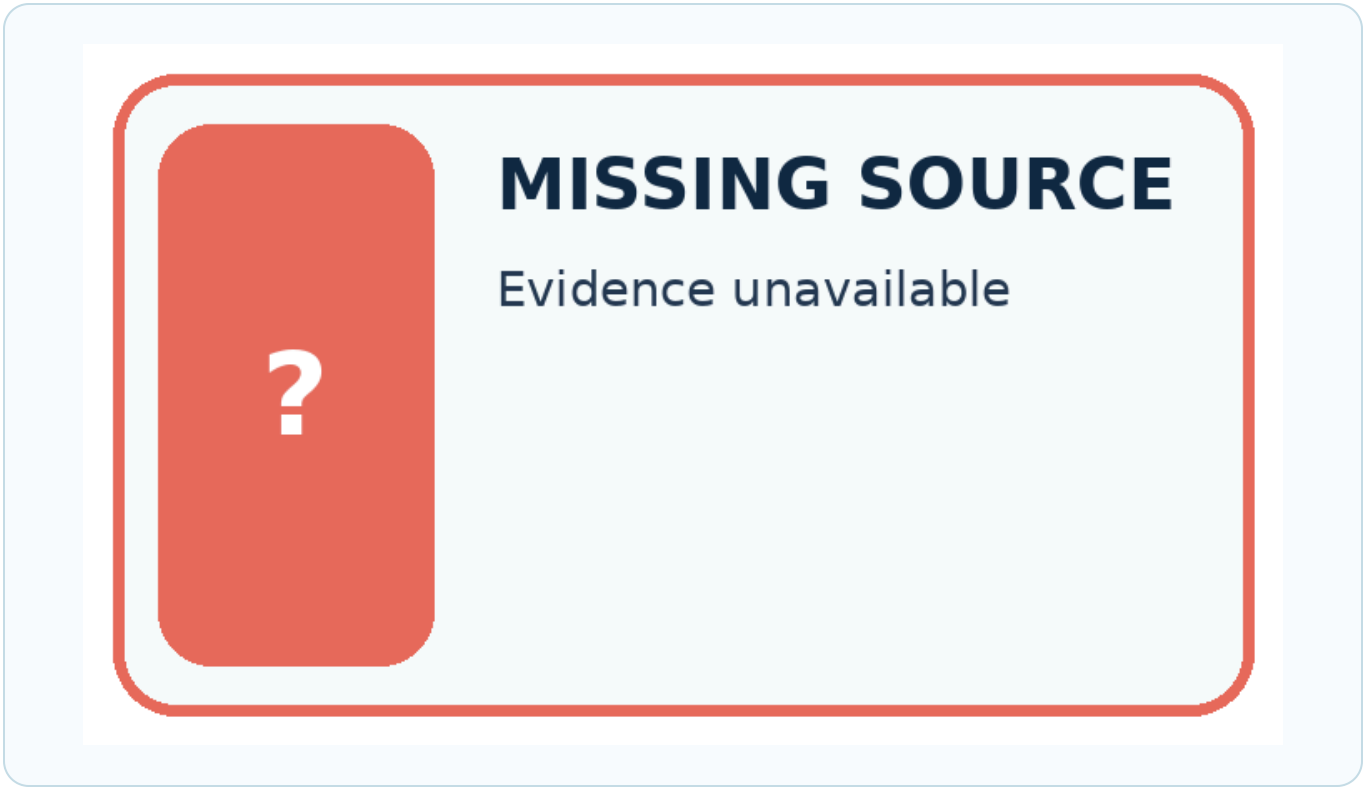
Four pictures with a purpose



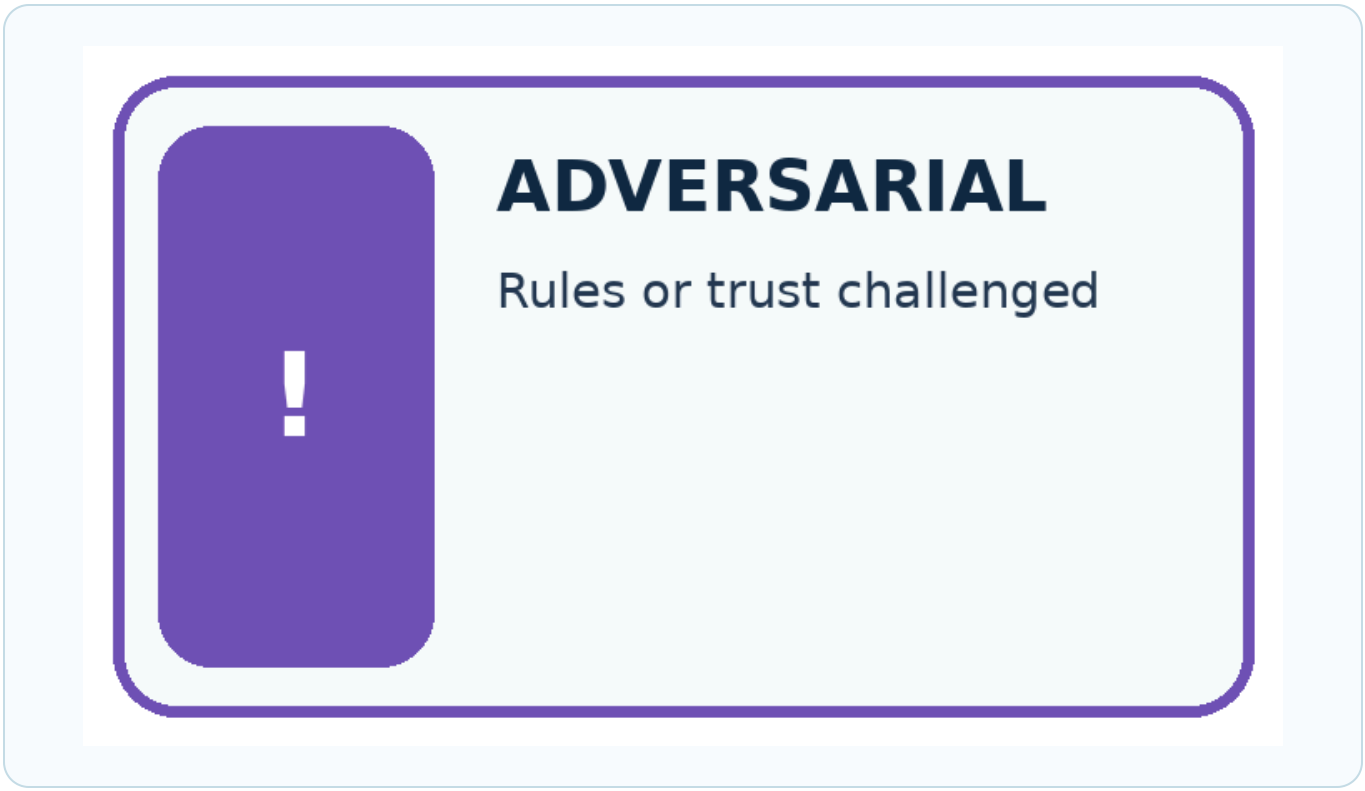
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Predict the supplied changed case

- Change: Remove the unsupported sentence while preserving the supported one.

▼ Check after predicting

Prediction: The revised one-sentence answer passes these two declared checks.

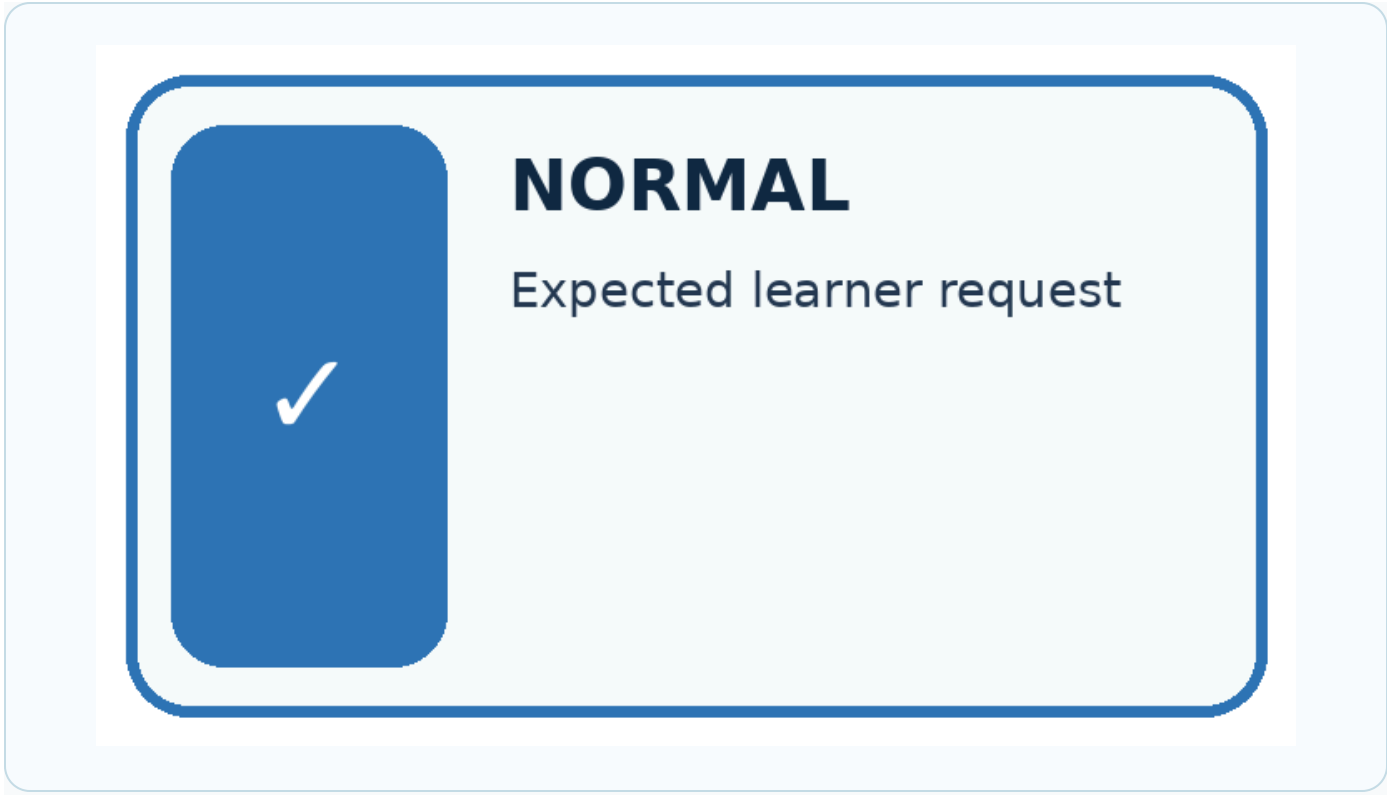
Reason: The remaining claim is supported and one sentence is within the maximum.

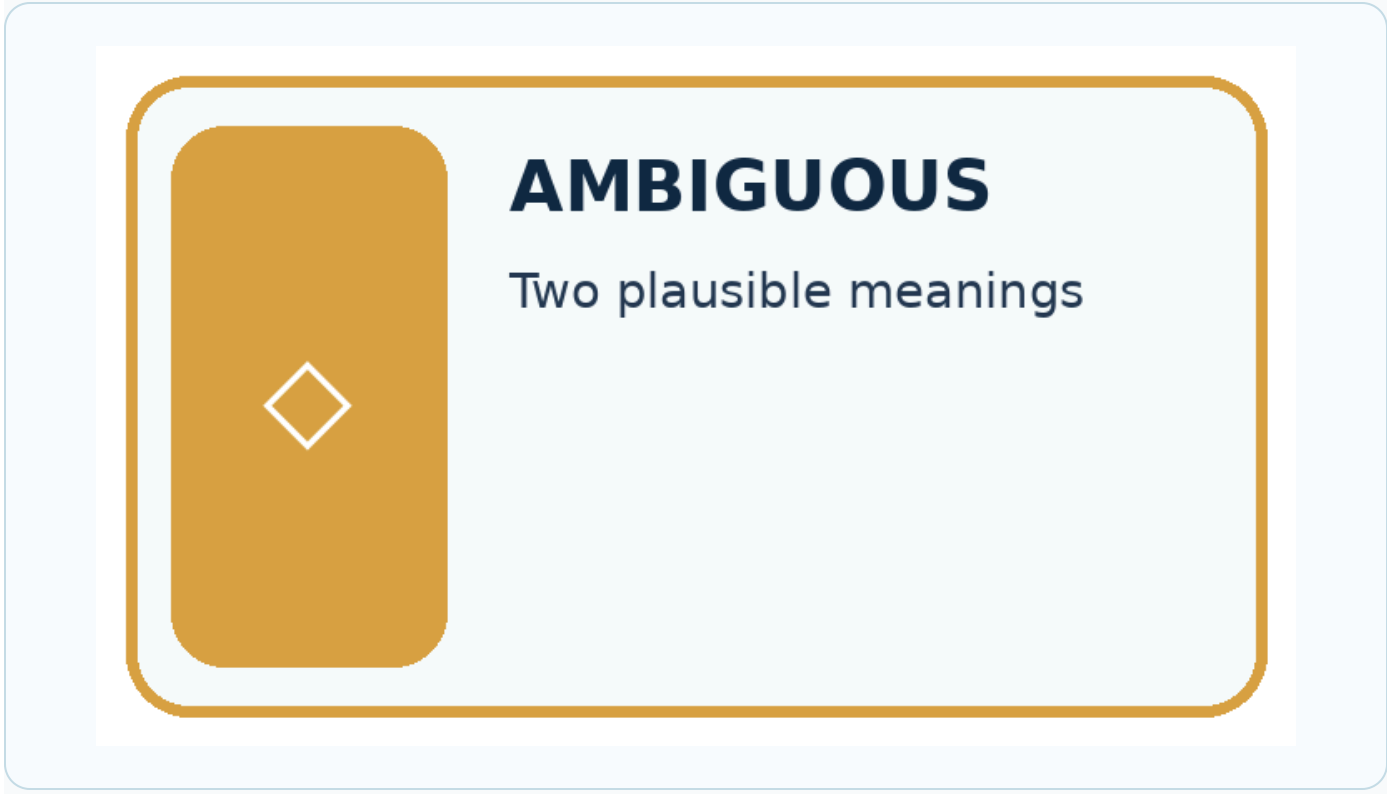
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace each criterion against S1 and the draft

> Removing unsupported content repairs the deciding defect.

Read the complete note, question and draft

Use a quality rubric whose safety gate cannot be traded for attractive formatting.

Approved S1 says: “A root **anchor**s a plant in soil.” Rubric: every claim supported by S1, at most two sentences. Draft: “A root **anchor**s a plant in soil. Roots grow exactly 2 cm each day.”

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

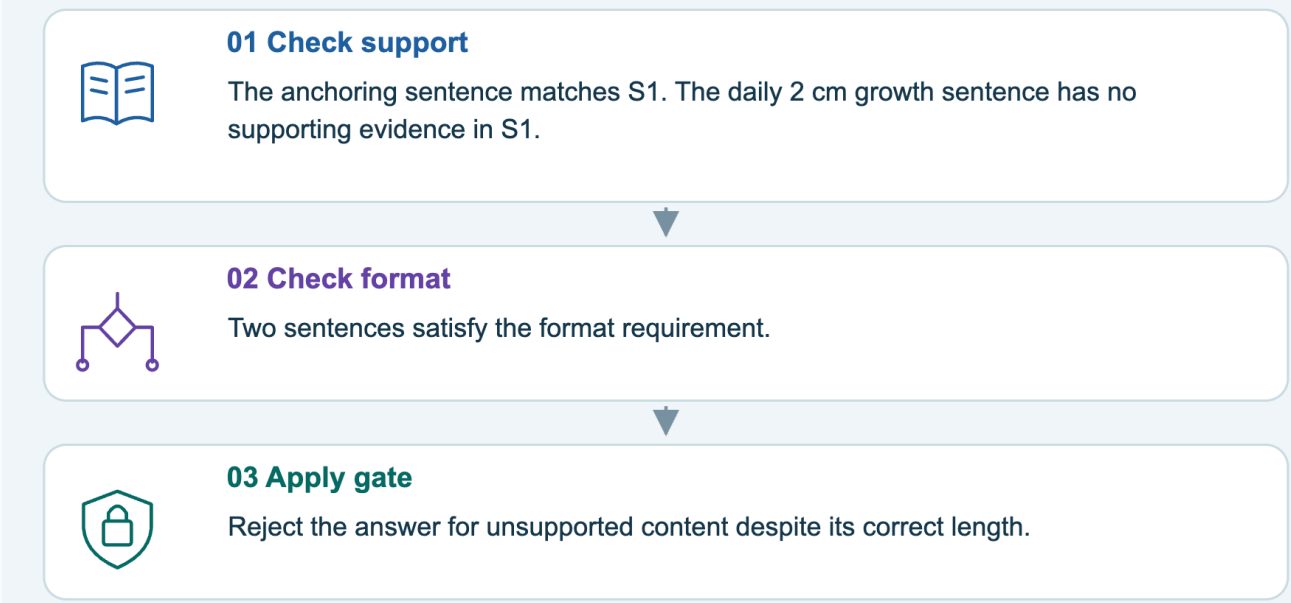
Current question: what role does a root have according to S1? S1 says only A root anchors a plant in soil. Draft has two sentences: the supported anchoring sentence and Roots grow exactly 2 cm each day. Under the three-criterion rubric, task fit and format pass, but source support fails. The source gives no daily growth measurement.

This is an evidence-support gate; do not claim that the exercise measures every dimension of safety.

Trace source support and format

AI ACADEMY / AI-3/25 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check support	The anchoring sentence matches S1. The daily 2 cm growth sentence has no supporting evidence in S1.
2. Check format	Two sentences satisfy the format requirement.
3. Apply gate	Reject the answer for unsupported content despite its correct length.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

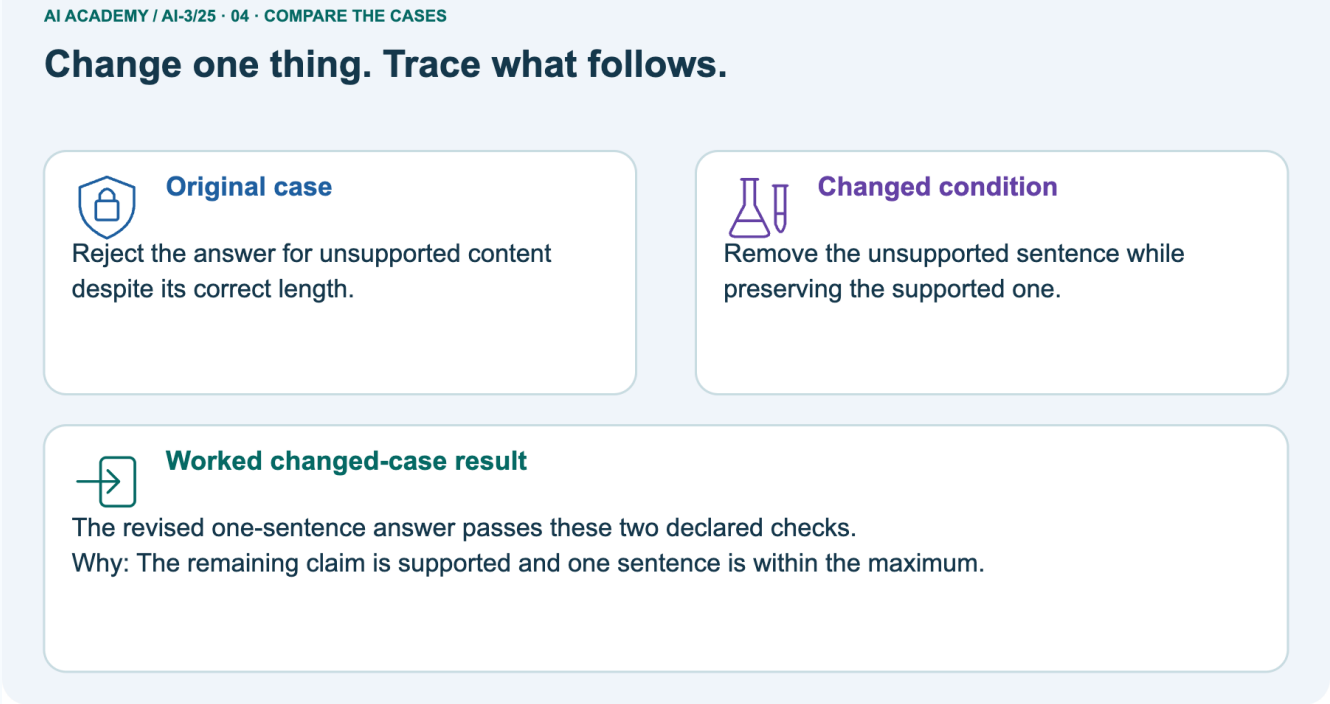
Compare each draft sentence directly with S1. The **anchor**ing claim matches the supplied source; the daily growth claim adds a measurement the source never gives.

Two sentences pass the format rule, but unsupported content fails the mandatory **evidence** gate. Removing that sentence repairs the support defect rather than merely polishing the wording.

Predict removing the unsupported sentence

- Change one thing: Remove the unsupported sentence while preserving the supported one. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State the small-rubric limit

- Passing this small rubric does not establish readability, completeness or benefit across every pupil and question.

- Read the labelled picture: Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Review · Separate evaluation-set fractions

Ten cases include six answerable and four missing-**evidence** cases. System answers five answerable correctly and abstains appropriately on three missing cases.

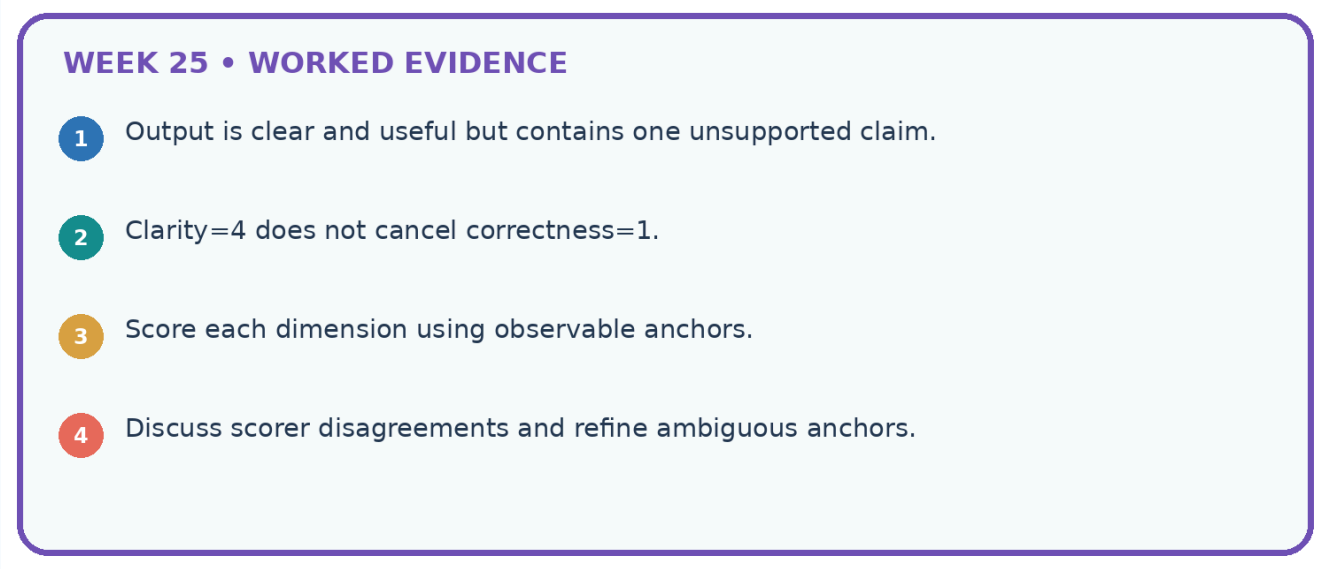
Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome. Different denominators answer different questions.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. **Output** is clear and useful but contains one unsupported claim.

2. Clarity=4 does not cancel correctness=1.

3. Score each **dimension** using observable **anchor**s.

4. Discuss scorer disagreements and refine ambiguous **anchor**s.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

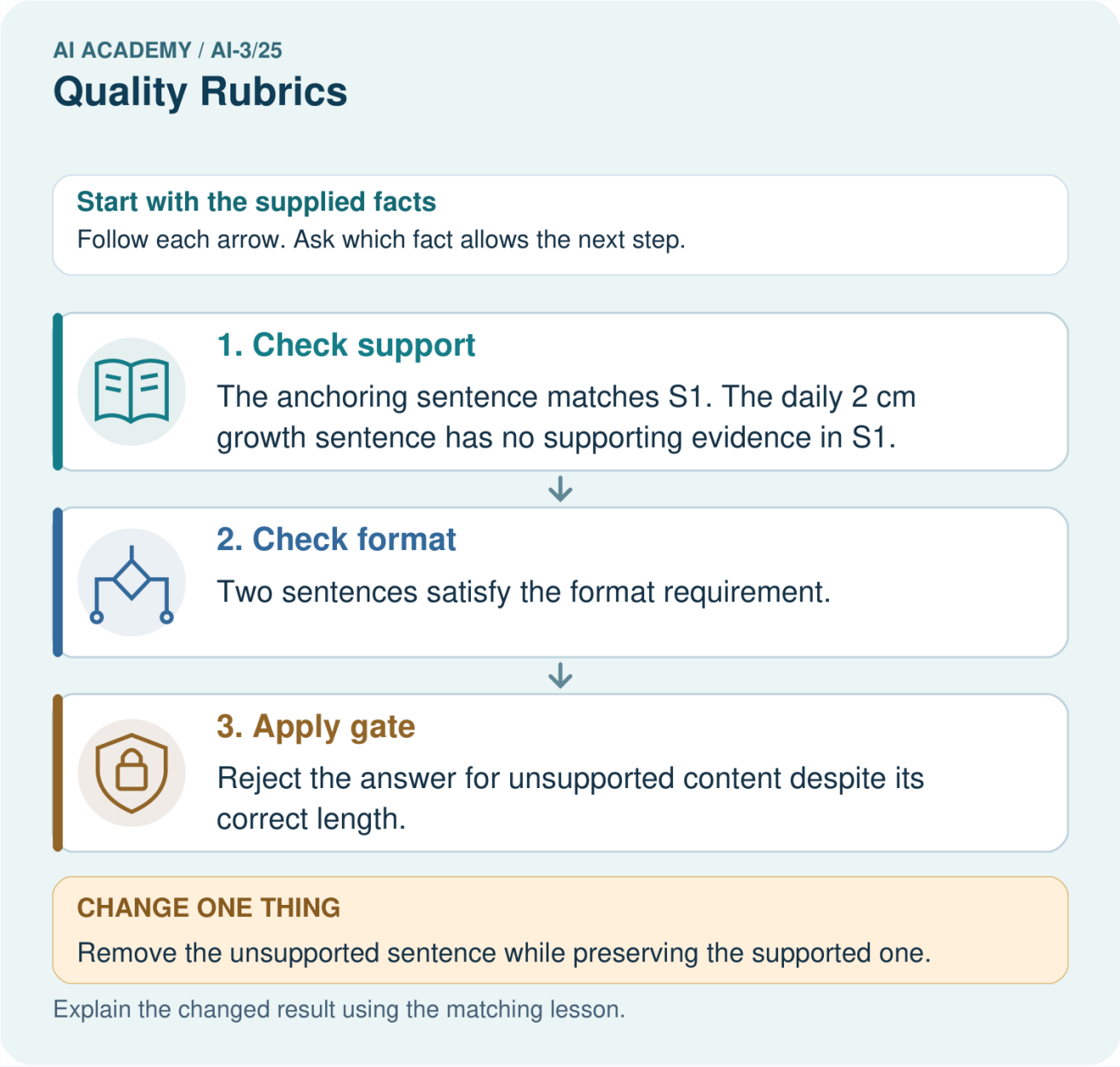
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Quality Rubrics

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a point total used to bypass a gate

> Repair the failed criterion and keep the original draft for comparison.

Repair the majority-of-checks shortcut

Claim to repair: Two of three checks passed, so release the answer. The declared rule requires all three, and factual support is mandatory. A count of passed criteria is descriptive; it does not override the release policy. A second shortcut is saying unsupported means false: the growth measurement is not established by S1, so omit or verify it rather than inventing proof that it never happens.

Design a revealing follow-up

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: One overall star rating explains product quality. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise criterion-by-criterion decisions with less help

> Write evidence before discussing the overall release decision.

Current task and response

- Use S1: A root anchors a plant in soil. Question: what role does a root have? Draft: A root anchors a plant in soil. Roots grow exactly 2 cm each day. Apply source support, task fit and at-most-two-sentence format separately, giving evidence for each. Apply the declared all-three release rule, then predict removing only the unsupported sentence.

My rubric decisions and evidence

Explain the pictured gate

- Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Explain the picture

My explanation using a pictured step and the given facts

Keep independent work separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Review · Earlier evaluation-set practice

Evaluation **coverage** limits the claim; prompts tuned on these cases need new reserved final cases.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “System evaluation sets”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Check your reasoning

Use the worked case for System evaluation sets. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

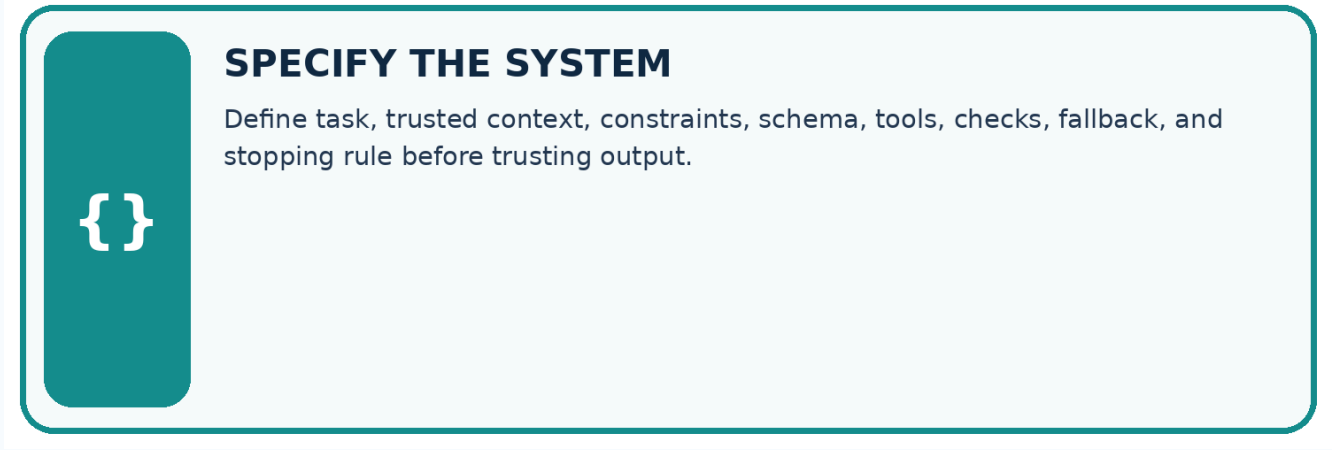
Answer on your own before discussion. If you are unsure, say which step you need help with.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Rate three complete outputs with the same rubric

A shared source and question make disagreements inspectable.

All supplied output cards

Output for the same root-role question and S1	Expected separate checks
A · A root anchors a plant in soil. Roots grow exactly 2 cm each day.	Support fail; task fit pass; format pass; release no.
B · A root anchors a plant in soil.	Support pass; task fit pass; format pass; release yes under these three checks.
C · A root anchors a plant in soil. A root anchors a plant in soil. A root anchors a plant in soil.	Support pass; task fit pass; format fail with three sentences; release no. Repetition can motivate a separately defined clarity criterion.

Rate, compare, justify and revise an anchor

1. Rate all three supplied outputs independently before uncovering or discussing expected checks. For every judgement cite an exact output clause, source passage or sentence count. Compare disagreements by dimension, not by total. If an anchor is ambiguous, write a more observable revision, version the rubric and rescore all outputs rather than moving the rule for one preferred draft.

   These are constructed output cards, not actual model evaluations. Passing B establishes only this small declared contract. Do not claim readability, full safety or learning benefit from these flags. Keep the new seed-coat assessment withheld.

Keep rating evidence and disagreements

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the three supplied root-role drafts and S1 with the declared source-support, task-fit and sentence-limit **anchor**s. Rate independently, justify each **dimension** and preserve the mandatory **evidence** gate. Revise ambiguous **anchor**s explicitly and rescore all **output**s; do not claim a live model or usability study.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: judge the fresh seed-coat draft

> Use only C2 and the stated criteria; retain your first attempt before feedback.

Current independent source, draft and response

- Approved C2 says: A seed coat protects the seed. Question: what role does the seed coat have? Draft: A seed coat protects the seed. Every seed opens exactly 48 hours later. Use the same three criteria: every factual claim supported by C2, answer the asked role question, at most two sentences. All three are required for release.

  Give a separate judgement with evidence for each, the release decision and one justified repair.

My rubric decisions and evidence

Feedback after the first attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Review · Earlier evaluation-rate tasks and responses**

Independent case: Apply a new case independently

Twelve cases: eight answerable, four unanswerable. Correct answers six; appropriate abstentions two. Compute separate and combined pass rates.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 25 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Two answers have equally polished language. One has three supported claims; the other has two supported claims and one invented claim. How should a quality rubric distinguish them?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Use a five-**dimension** rubric and block release when correctness or safety is below threshold.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

An answer earns 2 for format, 2 for tone and 0 for source support. It gets 4/6. Is it acceptable for a source-only assistant? Revise the rubric's decision rule.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's quality criteria reviewable

> Declare the release conditions before looking at preferred outputs.

This week's contribution

For the Verified Study Buddy, version the rubric alongside sources and outputs. Separate observable dimensions, define mandatory conditions and record a reason for every judgement. Keep a failed draft and repaired version. If adding clarity or usefulness criteria, supply observable anchors and record human disagreements rather than claiming an automatic score measures user benefit.

Broader quality requires appropriate evidence beyond this lesson's small rubric.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain the checkpoint and human choice

THIS WEEK'S CHECKPOINT Use a five-**dimension** rubric and block release when correctness or safety is below threshold.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

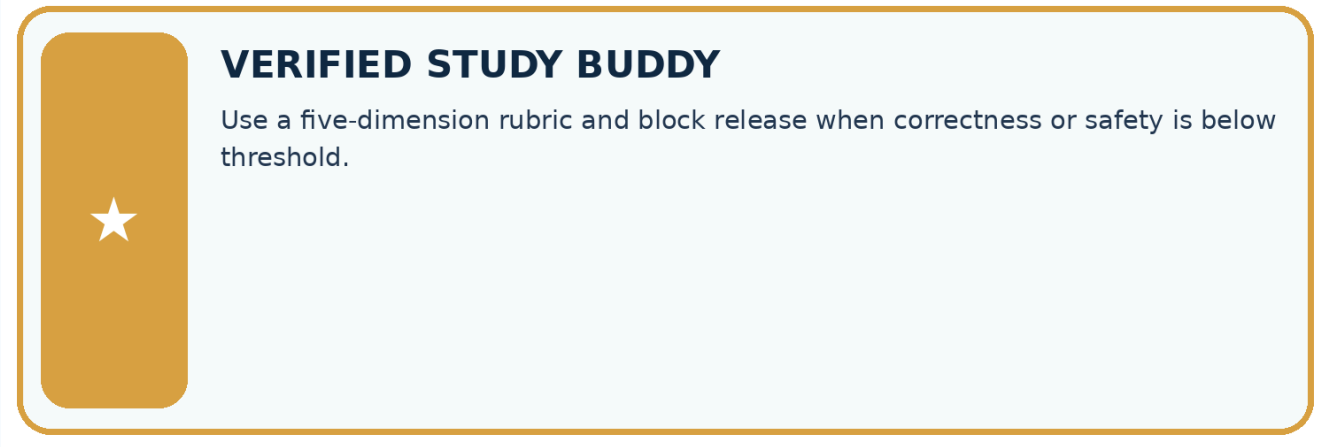
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the failed criterion and its evidence

> A count of passed checks cannot replace the declared release rule.

Explain without notes

- Close your notes. Explain why separate observable criteria and a mandatory evidence gate can produce a different decision from a total score.

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
dimension	One component or axis of a representation, such as one feature in a vector.	
anchor	A reference example used to make a judgment or scoring standard concrete.	
inter-rater agreement	The extent to which different labelers give matching judgments on the same cases.	
tradeoff	A situation where improving one desired property can make another harder to achieve.	

Terms for the connected explanation

Word	Meaning
Evaluation case	An input with evidence and prespecified expected behaviour

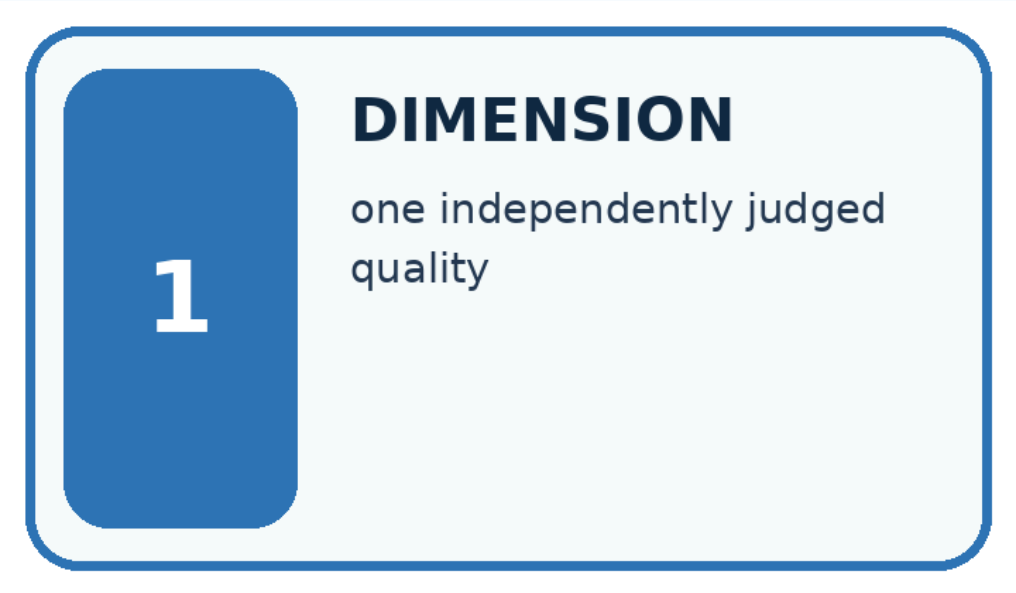
Word	Meaning
Coverage	Which relevant conditions are represented
Contract pass	Satisfying the declared requirements for one case

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



</details>


2
ANCHOR
description of a score level


3
INTER-RATER AGREE
how consistently evaluators score



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

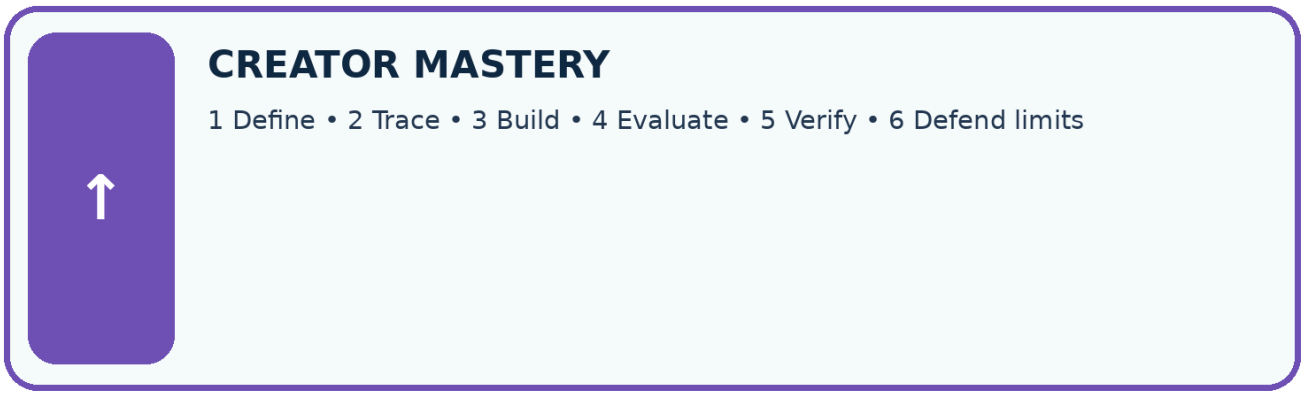
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Judge the new seed-coat draft independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

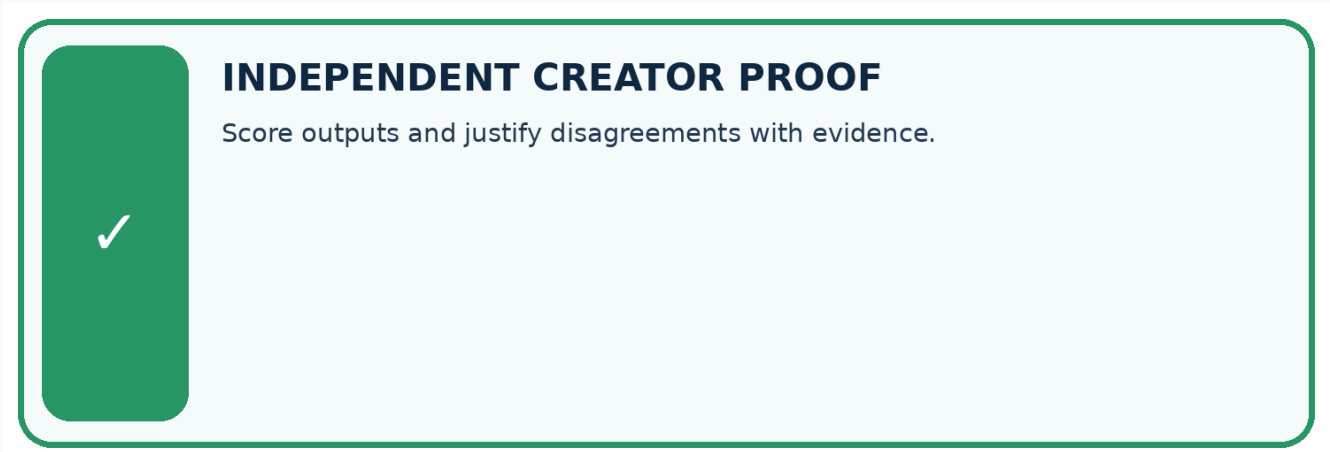
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer ‘why?’

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week’s idea

Use after practice. Judge the new seed-coat draft independently is the selected independent case.



EXPLANATION CHECK Require separate criterion judgements with direct **evidence** and the declared release rule.

Write or draw your reasoning and evidence.

▼ **Extra practice: A total can hide a required failure**

REFERENCE · SECTION 13

Extra practice: A total can hide a required failure

> Try the question before opening its worked explanation.

Read the supplied case and question

Rubric gives 0–2 each for support, usefulness and safety. Response is useful and safe but unsupported.

Why can a total mislead?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

It may get 4/6 while failing essential support. Declare mandatory criteria before testing so other points cannot compensate.

Evidence or stage	Value or result
Support	0
Usefulness	2
Safety	2

▼ Earlier wording, alternatives and picture transcripts

Week 25 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A total can hide a required failure

Rubric gives 0–2 each for support, usefulness and safety. Response is useful and safe but unsupported.

Why can a total mislead?

It may get 4/6 while failing essential support. Declare mandatory criteria before testing so other points cannot compensate.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of system evaluation sets	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Use a quality rubric whose safety gate cannot be traded for attractive formatting.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Use a quality rubric whose safety gate cannot be traded for attractive formatting.
- Which labelled step matches this detail? Reject the answer for unsupported content despite its correct length.
- 1. Check format
- 2. Apply gate
- 3. Check support
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Apply gate Reject the answer for unsupported content despite its correct length. Compare each draft sentence directly with S1.

  The **anchor**ing claim matches the supplied source; the daily growth claim adds a measurement the source never gives. Two sentences pass the format rule, but unsupported content fails the mandatory **evidence** gate.

  Removing that sentence repairs the support defect rather than merely polishing the wording.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: Remove the unsupported sentence while preserving the supported one. Which conclusion is supported by the supplied facts?
- 1. The revised one-sentence answer passes these two declared checks.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: The revised one-sentence answer passes these two declared checks. The remaining claim is supported and one sentence is within the maximum.

My selected conclusion, deciding fact and explanation

▼ Check the changed case

Repair a missing building block

Repair and save your work

Review Prototype a Grounded Assistant (ai-3/24). Goal: Explain the **mechanism** and terminology of capstone grounded assistant. Short practice: Can PL2 alone support “check the soil daily”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Requests Responses and JSON (ai-3/22). Goal: Parse JSON into Python records. Short practice: Would converting the text silently prove the source count is correct?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review User Need and AI Contribution (ai-3/19). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What non-AI baseline would reveal whether this helper adds value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

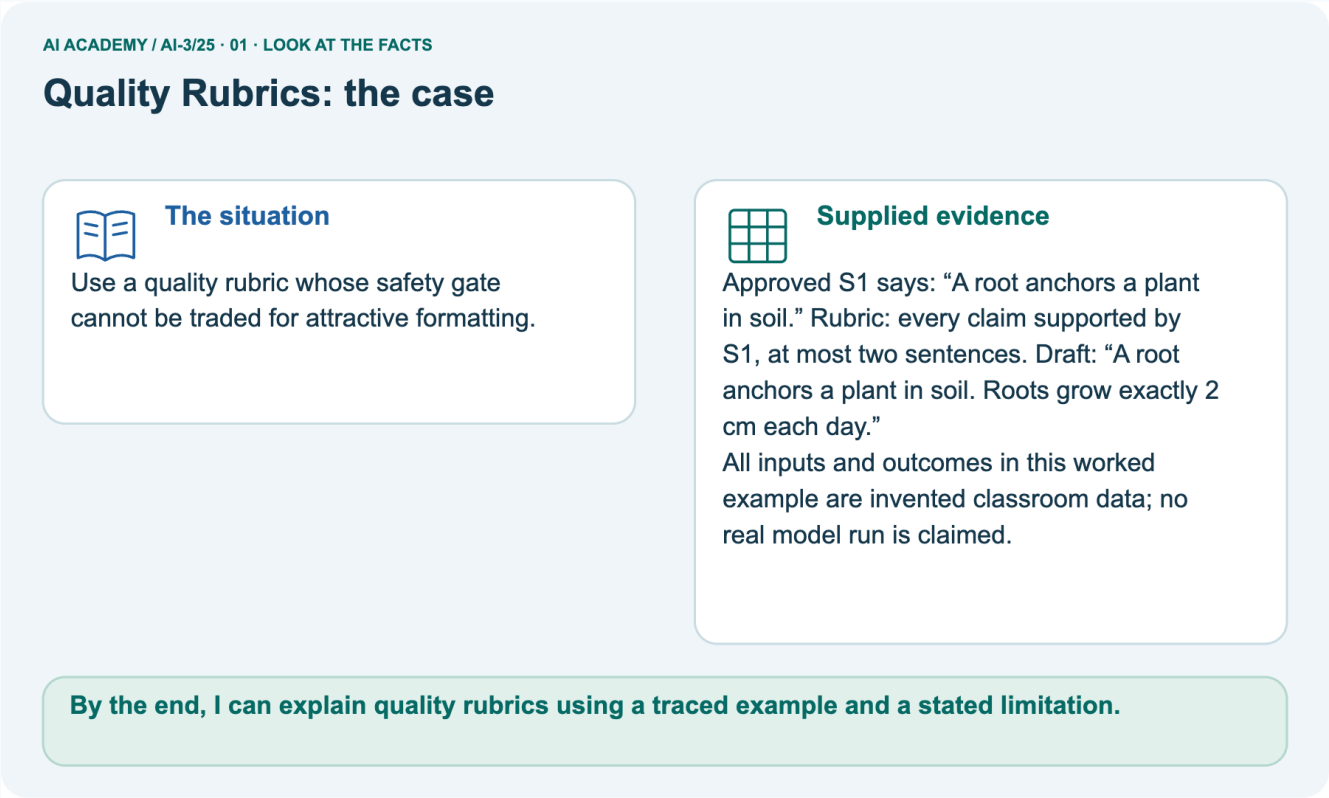
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Quality Rubrics: the case



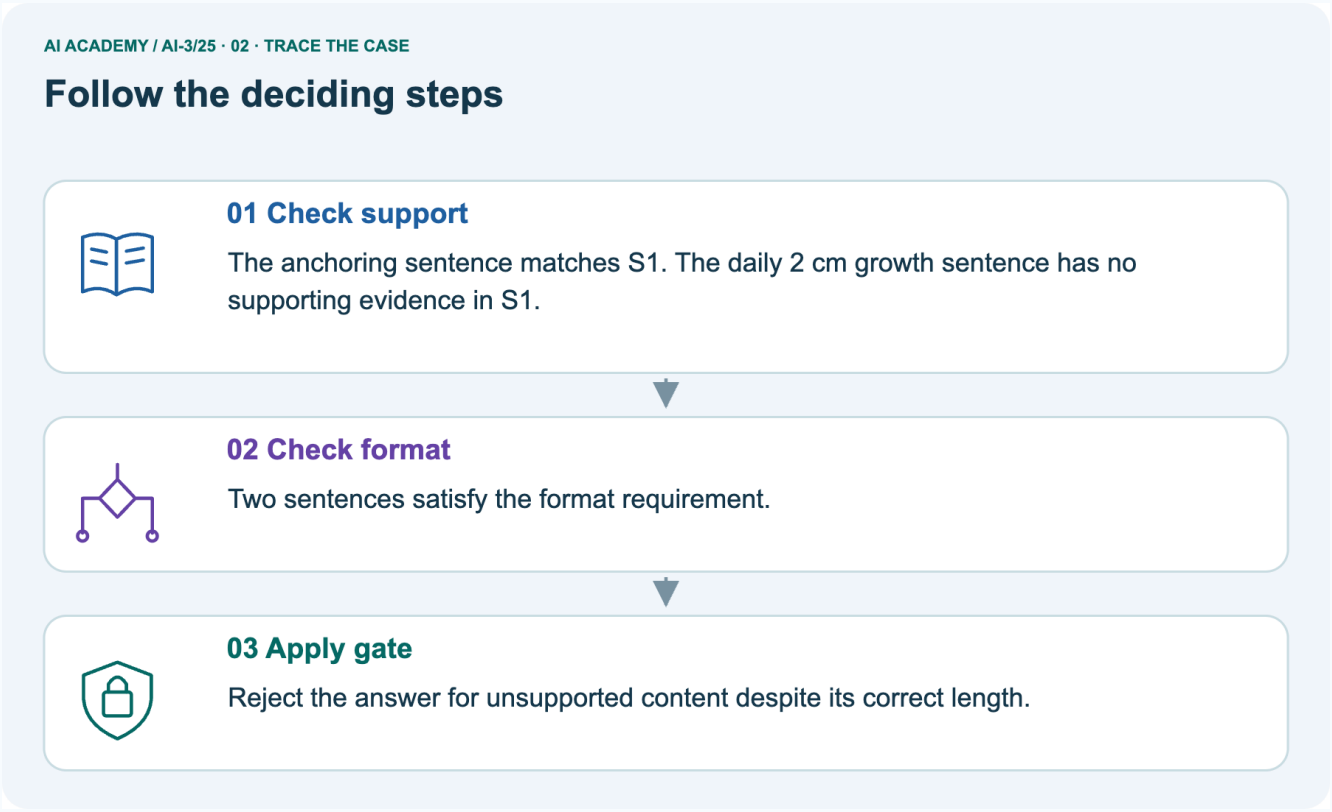
Follow the labelled picture with the matching explanation. Open the vector diagram

Use a quality rubric whose safety gate cannot be traded for attractive formatting. Supplied facts: Approved S1 says: “A root **anchor**s a plant in soil.” Rubric: every claim supported by S1, at most two sentences.

Draft: “A root **anchor**s a plant in soil. Roots grow exactly 2 cm each day.” All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



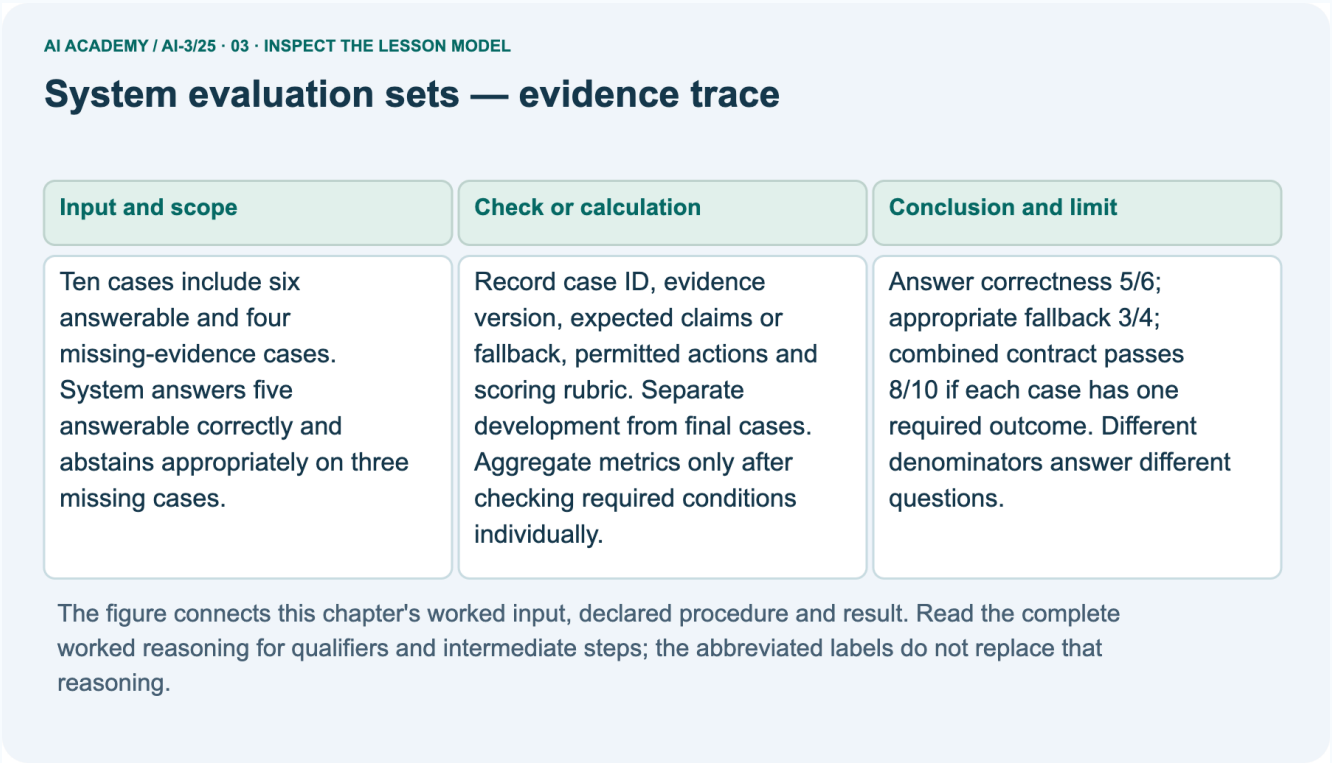
Follow the labelled picture with the matching explanation. Open the vector diagram

Check support: The **anchor**ing sentence matches S1.

The daily 2 cm growth sentence has no supporting **evidence** in S1. → Check format: Two sentences satisfy the format requirement. → Apply gate: Reject the answer for unsupported content despite its correct length.

Point to each numbered stage and explain which recorded **input** it uses.

03. System evaluation sets — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

System evaluation sets — **evidence** trace. Ten cases include six answerable and four missing-**evidence** cases.

System answers five answerable correctly and abstains appropriately on three missing cases. | Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric.

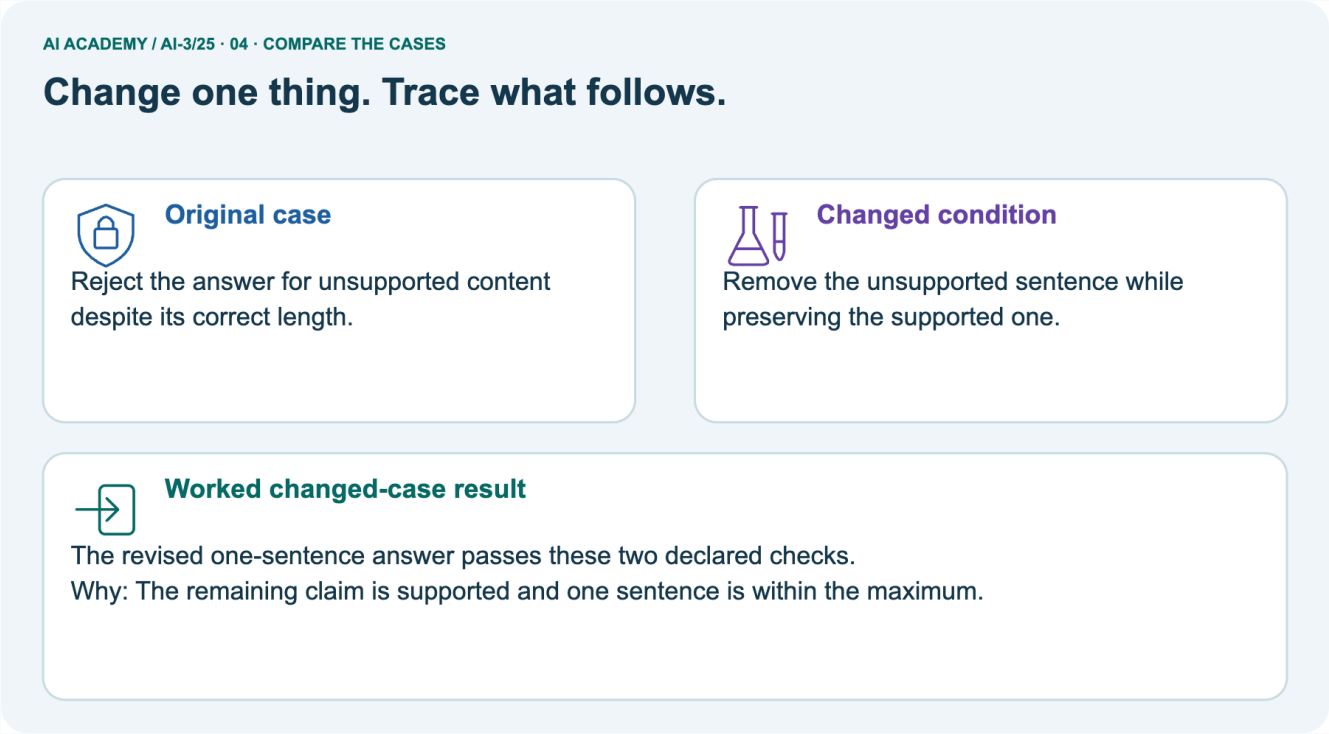
Separate development from final cases.

Aggregate metrics only after checking required conditions individually. | Answer correctness 5/6; appropriate fallback 3/4; combined **contract pass**es 8/10 if each case has one required outcome.

Different denominators answer different questions.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



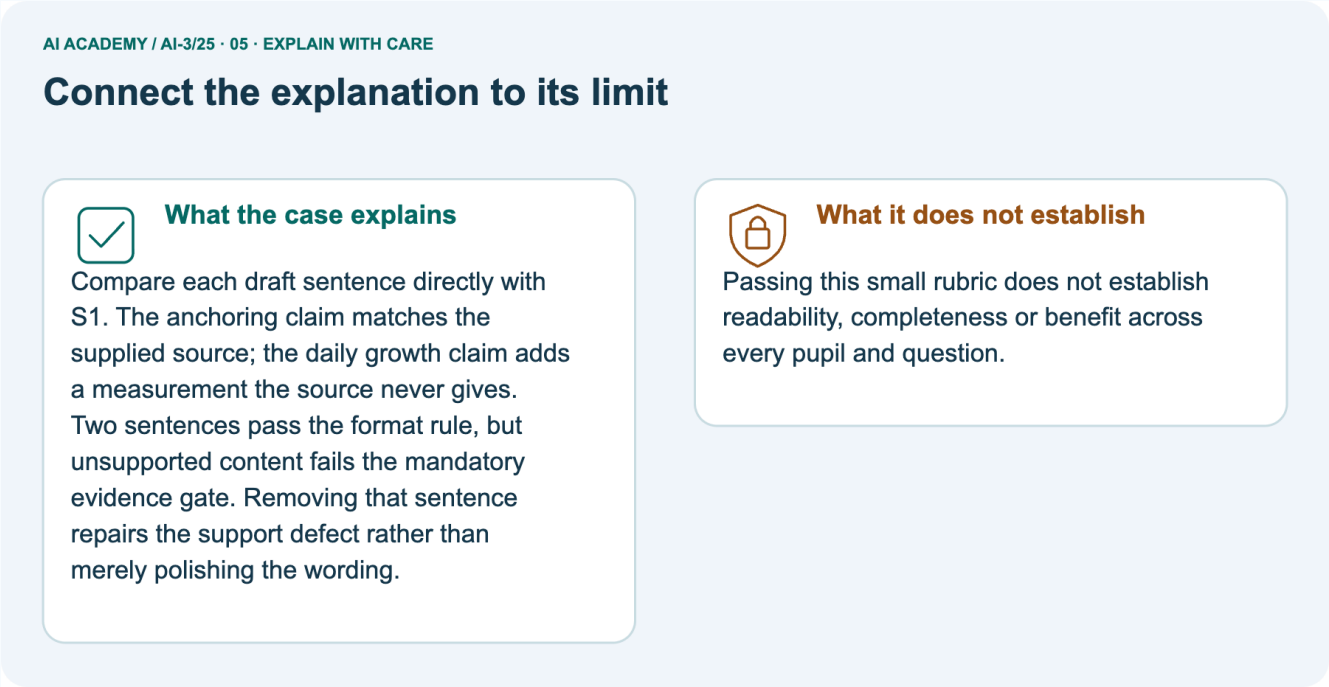
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Reject the answer for unsupported content despite its correct length. Changed condition: Remove the unsupported sentence while preserving the supported one.

Expected result: The revised one-sentence answer passes these two declared checks.
Reason: The remaining claim is supported and one sentence is within the maximum.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

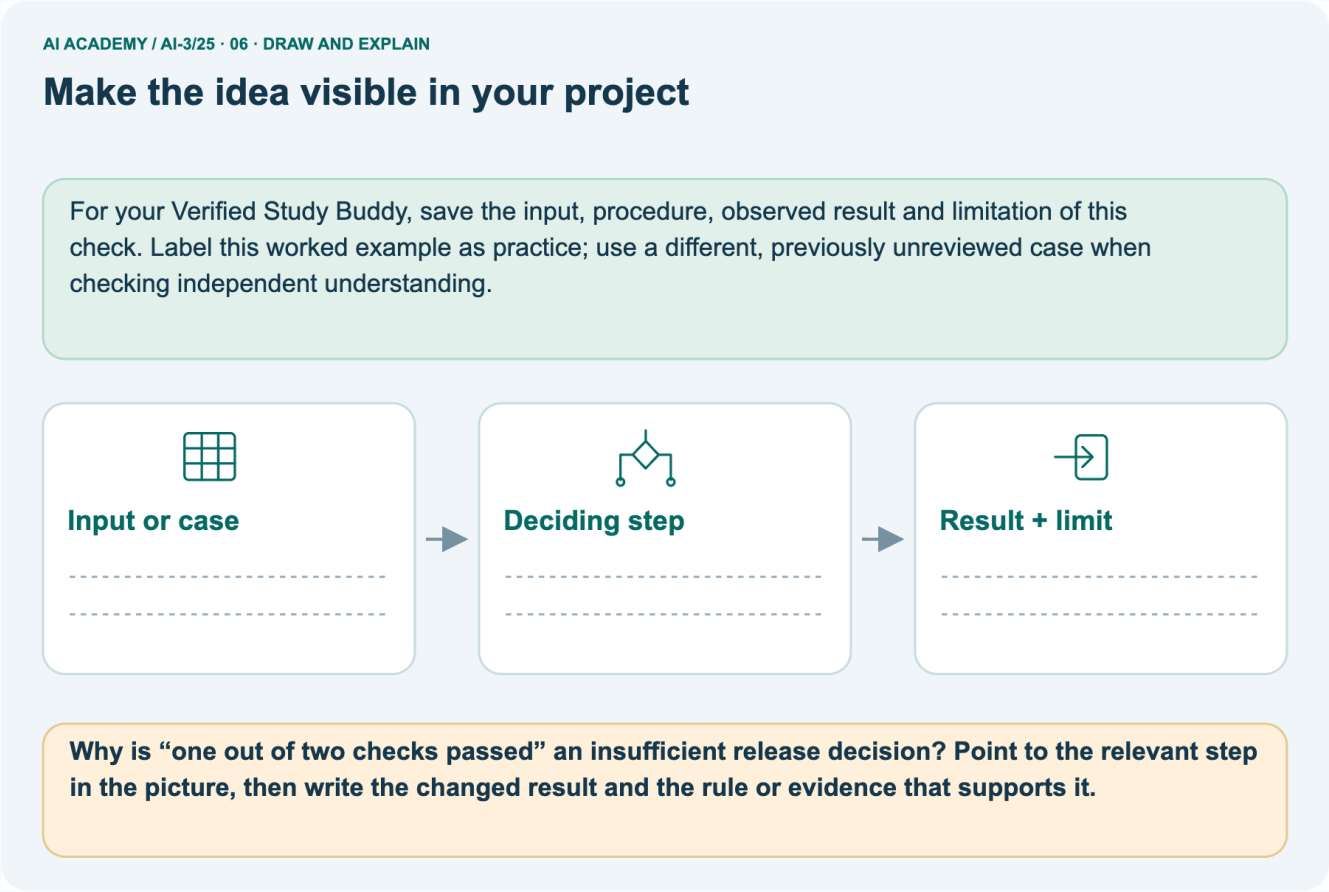
Compare each draft sentence directly with S1. The **anchor**ing claim matches the supplied source; the daily growth claim adds a measurement the source never gives.

Two sentences pass the format rule, but unsupported content fails the mandatory **evidence** gate. Removing that sentence repairs the support defect rather than merely polishing the wording.

Limit: Passing this small rubric does not establish readability, completeness or benefit across every pupil and question.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Quality Rubrics: the case

02. Follow the deciding steps

03. System evaluation sets — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.